

GUIDELINES For VIRTUAL REALITY: Think, Explore And Create

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This article provides an overview of the best practices that must be followed for the development of VR

The virtual reality (VR) revolution is bigger than the advent of the cell phone. Sound, touch, depth and emotion is all integral to the VR experience. Emergence of virtual theme parks have made a positive impact on the VR gaming industry. Oriental Science Fiction Valley in Guizhou Province, opened in 2017, is one such setup. Designing an experience that is comfortable for the users is the most important consideration. Ignoring fundamental best practices in VR experiences can lead to discomfort. Therefore it is important to follow the best practices or solutions to minimise discomfort.

ISO 9241-391:2016 standard for image content reduces the risk of photosensitive seizures. VR requires new ways of thinking about space, dimension, interaction, navigation and immersion.

Main challenges with VR design are balancing immersion, usability and comfort. This article provides the foundation to build a comfortable and engaging VR experience. It will help you create experiences that will transport users to whole new worlds. Note that, excessive use of VR without taking breaks is not recommended for developers and end-users alike.

Ground design principles

Poorly-designed VR experiences can cause motion sickness. Like physical reality, the ground-to-horizon relationship is important in VR. Ground can be an open or cluttered environment. A cluttered environment allows locomotion only at openings, whereas an open environment allows loco-

motion in any direction over the ground.

Conventional wayfinding user interface (UI) elements helps users with navigation via a certain path in a cluttered environment. Parallax mapping shifts the texture coordinates of the sampled surface using an additional height map. It does not affect the collision surface and is best utilised on surfaces with fine details.

Input methods

Controllers allows users to pick up and interact with other objects in a VR experience. Controllers for platforms need different best practices and capabilities. Mobile device controllers have three degrees of freedom (3DOF), while Rift controllers have six degrees of freedom (6DOF).

3DOF controllers track orientation of the controller, but do not track its position in space. 6DOF controllers support both orientation and positional tracking. 6DOF enables virtual hands that can interact with the VR environment in devices.

Role of sound and light

It is important to understand scale while designing VR. Tiny spaces may make people claustrophobic, while huge spaces may lead them to get lost.

There are many ways to guide people through a virtual environment. Light can be used to reveal a path to follow and audio can be used for spatial positioning. Flicker is the rapid change of lightness and darkness on screen. For areas outside the centre of players' viewpoint, using darker colours as brighter screen images produce more flicker. The higher the refresh rate, the lower the perceptible flicker.

A gentle introduction to the new environment can be achieved by changing soundscape at first, then the image. This lowers the shock factor by building a mental image of the environment via sound. Sound is a great way to provide feedback when users touch objects. Holophonic sound tells if the sound is coming from above, below or behind them.

3D rendering of man wearing virtual reality glasses surrounded by virtual data

